Lewis Structures

Answer key with worked solutions

A. Counting electrons

1. Valence electrons.

> (a) 4 + 4 × 1 = **8** (b) 5 + 3 × 7 = **26** (c) 5 + 4 × 1 − 1 = **8** (d) 4 + 3 × 6 + 2 = **24**

B. Drawing structures

2. H_2S, NF_3, CCl_4.

> (a) 8 e^-: H–S–H, **2 lone pairs on S** (b) 26 e^-: N bonded to three F, **1 lone pair on N**, 3 lone pairs on each F (c) 32 e^-: C bonded to four Cl, no lone pairs on C, **3 lone pairs on each Cl**

3. CH_2O and C_2H_4.

> (a) 12 e^-: C single-bonded to two H and **double-bonded to O**; 2 lone pairs on O. (b) 12 e^-: **C=C double bond**, each C single-bonded to two H; no lone pairs.

4. CS_2.

> 16 e^- (like CO_2): **S=C=S**, with 2 lone pairs on each S: **4 lone pairs** in total.

5. NH_4^+.

> 8 e^-: N bonded to four H, no lone pairs, drawn in square brackets with a + charge. Formal charge on N = 5 − 0 − ½(8) = **+1** (equal to the ion's charge).

C. Formal charge and resonance

6. Ozone.

> O=O–O: the central O has 1 lone pair, the double-bonded O has 2 lone pairs and the single-bonded O has 3 lone pairs. Formal charges: double-bonded O 6 − 4 − 2 = **0**; central O 6 − 2 − 3 = **+1**; single-bonded O 6 − 6 − 1 = **−1**. They add up to 0, the charge of the molecule.

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7. Nitrate.

> 24 e^-: N in the centre with **one N=O double bond and two N–O single bonds**. The three resonance structures differ only in which O has the double bond; they are linked by ↔ and drawn in brackets with a − charge. Formal charges: N +1, each single-bonded O −1, double-bonded O 0. Because the real ion is a hybrid, **all three N–O bonds are the same length**, between a single and a double bond.

D. Exceptions

8. BF_3.

> Boron has only 3 valence electrons and forms three single B–F bonds, so it is surrounded by only **6 electrons**, an incomplete octet.

E. Challenge

9. XeF_4.

> Xe is bonded to four F, each F with 3 lone pairs (8 + 24 = 32 e^-). The remaining 4 e^- go on Xe as **2 lone pairs**. Xenon is surrounded by 8 + 4 = **12 electrons**, an expanded octet. This is possible because xenon is a period-5 element, large enough to hold more than 8 electrons in its valence shell.

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